

## CSCE 638 NATURAL LANGUAGE PROCESSING · MODULE 07 · LECTURE NOTES

# Pretraining and Transfer

*Why training on everything beats training on your task.*

## MODULE OUTCOMES

- Explain the pretraining objectives and what each suits.
- Explain why transfer works unusually well here.
- Explain contextual representations and what they fixed.
- Explain scaling and what it does and does not deliver.
- Choose a pretrained model for a task.

## 1 The objectives

| Objective                 | The task                                                                   | What it suits                                                                                       |
|---------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Causal language modelling | Predict the next token from the left context only.                         | Generation; decoder-only architectures (Module 06 §3's causal mask).                                |
| Masked language modelling | Predict tokens that have been masked out, using the context on both sides. | Understanding tasks; encoder-only architectures.                                                    |
| Span corruption           | Reconstruct spans that have been deleted from the input.                   | Both; encoder-decoder architectures — the attempt to have bidirectionality and generation together. |

Causal models can generate and can see only the left context; masked models see both sides and cannot generate — which is the fundamental trade, and span corruption is the attempt to have both at some cost in efficiency. The bidirectional-versus-generative distinction is the organising fact of this table, and it follows directly from Module 06 §3's mask.

**THE OBJECTIVE IS A MEANS, AND THE REPRESENTATIONS ARE THE PRODUCT**

Nobody actually wants a model that fills in masked words. The objective exists because solving it requires learning syntax, semantics, and a great deal of world knowledge — which is Module 01 §3's argument, and the objective is the instrument rather than the goal.

So the useful output is the internal representations — and the measure of a pretraining objective is how useful the representations it produces turn out to be, not how well the objective itself is solved. A model that solved masked prediction perfectly by some shortcut would be worthless.

Which explains the apparently arbitrary design choices: masked language modelling's 15% masking rate, the replace-with-random-token trick, and the span length distributions are all tuned for representation quality rather than for the proxy task's score.

And it explains why causal models came to dominate: the objective is harder, every token serves as a prediction target rather than only the masked 15%, and the resulting model can also generate — which turned out to matter more in practice than bidirectionality did, and was not obvious in advance.

## 2 Contextual representations

**A PRETRAINED MODEL GIVES A DIFFERENT VECTOR FOR EACH OCCURRENCE OF A WORD**

'Bank' in a financial sentence and 'bank' beside a river receive genuinely different representations — because the representation is computed from the whole sentence rather than looked up by word identity, which is the entire architectural point of Module 06's self-attention.

Which is Module 04 §3's decisive limitation removed outright — and it is why pretrained models improved essentially every task at once rather than one task at a time, which is what made the transition so abrupt.

And the layers differ systematically in what they encode: lower layers carry more surface and syntactic information, and higher layers more semantic and task-relevant information — a finding that has held up robustly across many probing studies and several model families.

So which layer to use is a real decision — and 'the last one' is frequently not the best answer for a task that depends on syntax or on surface form, where a middle layer measurably outperforms it.

Concatenating several layers is often better still, and costs nothing but memory.

### 3 Why transfer works unusually well here

- **The pretraining data covers the target distribution.** Nearly any text task involves text that resembles the pretraining text — which is unusually true in language and is **the main reason transfer works so well here**, as opposed to in settings where the target data looks nothing like anything available at scale.
- **The required knowledge is genuinely shared.** Syntax, word senses, and factual knowledge are needed by **essentially every task**, so learning them once serves all of them — whereas in a narrower domain the useful knowledge may be task-specific.
- **The objective is genuinely hard**, so it forces rich representations rather than admitting a shortcut — **which is not true of every self-supervised objective**, several of which turned out to be solvable by exploiting a surface artefact.
- **And the scale is available.** Text is abundant in a way that labelled examples simply are not (Module 01 §3), so the pretraining can be done at a scale the fine-tuning never could.
- **So the gain is largest where labelled data is smallest** — **which describes most real applications**, and is why this changed practice rather than only benchmarks. **The distribution-coverage reason is the one that does not transfer to other domains**, and is why pretraining has worked so much less dramatically in tabular settings (CSCE 633).

#### SCALING WORKS, AND IT IS WORTH BEING PRECISE ABOUT WHAT IT DELIVERS

**Loss falls predictably with parameters, data, and compute** — a smooth power law holding over several orders of magnitude, **which is an unusually clean empirical result** for machine learning and is what made large investments forecastable.

**And the compute-optimal allocation matters a great deal: the early very large models were substantially undertrained**, and **training a smaller model on considerably more data turned out to be better for the same compute budget** — a correction that changed how models are sized.

**What scaling delivers is lower loss, and capability follows imperfectly. Some abilities improve smoothly with scale, and some appear to improve sharply** — though **the apparent sharpness depends substantially on the choice of metric**, since a discontinuous metric over a continuously improving model produces an apparent jump (Module 10 §1).

**And what it does not deliver is reliability. Larger models are more capable and still fabricate** (Module 12 §3) — **so scale moves the average and not the worst case**, which is the distinction that matters for anything deployed.

### 4 Choosing one, and what it does not give you

- **Start from the task shape.** Classification or extraction favours an encoder; generation requires a decoder; sequence-to-sequence transformation suits an encoder-decoder — and the mismatch is expensive rather than impossible.

- **Then the domain. A model pretrained on text like yours beats a larger general-purpose one** on specialised vocabulary — clinical, legal, and source code especially, where the tokenisation alone makes a large difference (Module 02 §3).
- **Then the constraints:** latency budget, memory, cost per request at your expected volume, and **whether the data is permitted to leave your infrastructure** — which is frequently the binding constraint and is a governance question rather than a technical one.
- **Then the licence and the provenance**, which is a real constraint and **is routinely discovered late**, after the system is built — and which Module 13 treats as part of an honest claim.
- **And start with the smallest model that could possibly work**, because **iteration speed matters more than capability while you are still learning what the task needs. This is the advice most often ignored**, and **a fast feedback loop finds the data problems that a slow one hides** — which is where most of the real difficulty in these projects turns out to live.

#### AND WHAT PRETRAINING DOES NOT GIVE YOU

**Knowledge of your specific data** — your schema, your customers, your product's terminology, your internal documents, and anything at all that happened after the training cutoff (Module 11's whole subject).

**Reliable factuality. The model learned what text looks like, and that includes what *false* text looks like** — and **it has no mechanism that separates the two**, because the training objective did not distinguish them (Module 03 §1's last point).

**Alignment with your task's definition. Your notion of a 'relevant' document, a 'serious' complaint, or an 'acceptable' summary is not in the pretraining data** — which is exactly what Module 08's adaptation supplies, and why some adaptation is nearly always necessary.

**And freedom from the corpus's biases**, which it represents faithfully — **Module 04 §3's sentence applies unchanged at this scale**, and Module 12 §2 is where the consequences are treated.

## Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

| Resource                                                                | What it covers for this module                                                                        |
|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Devlin et al., and Radford et al. — BERT and the GPT series (free)      | <b>§1's two objectives in the originals</b> — read them together, because the contrast is the lesson. |
| Hoffmann et al. — Training Compute-Optimal Large Language Models (free) | <b>§3's allocation correction</b> — the result that established the earlier models were undertrained. |
| Tenney et al. — BERT Rediscovered the Classical NLP Pipeline (free)     | <b>§2's layer finding</b> — and a satisfying connection to Module 01 §2's pipeline critique.          |

| Resource                                                   | What it covers for this module                                                                               |
|------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Schaeffer et al. — Are Emergent Abilities a Mirage? (free) | §3's <b>sharpness caveat</b> — the argument that the metric produces the discontinuity. Read with Module 10. |

## Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **State the three objectives** and what each architecture suits.
- 2 **Explain why 15% masking** is a representation-quality decision rather than a task one.
- 3 **Extract representations for an ambiguous word** in two sentences, and measure their distance.
- 4 **Compare that to the static embedding distance** from Module 04.
- 5 **Probe three layers** for a syntactic property and for a semantic one, and report which layer wins.
- 6 **List the four reasons transfer works here** and say which does not generalise.
- 7 **Find a scaling curve** and identify the power law region.
- 8 **Construct a metric that makes smooth improvement look sharp.**
- 9 **Choose a pretrained model for your Project 2 task** and justify it on all five criteria.
- 10 **List four things pretraining did not give you** for that task.

## Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Name the three objectives and the trade between the first two.
- 2 Why is the objective a means rather than the goal?
- 3 Why did causal models come to dominate?
- 4 What limitation do contextual representations remove?
- 5 What do the layers differ in, and what follows for layer selection?
- 6 Give four reasons transfer works here, and which is language-specific.
- 7 What does scaling deliver predictably?
- 8 Why might apparent sharp capability gains be a metric artefact?
- 9 What does scale not deliver?
- 10 Give five criteria for choosing a model, and four things pretraining omits.